

PIPING ARRANGEMENT EDIT

ACE Arrangement Edit (E3D)

E3D arrangement (plan) 200-series pipeline — centerline, symbols, line-no stretch, dimensioning, tag-no arrange, with option dialog.

Key Purpose

Why this program

1. Auto-finish raw E3D arrangement DXF into an issue-ready plan

– One Apply runs the full 200-series chain — centerline, symbols, line-no stretch, dimensioning, tag arrange

→ **Effect:** Hours of manual drafting cleanup per plan collapse to a single click, with consistent output every time

2. Read the drawing and pre-set the options for you

– Scans layers/text to detect elevation mode, drawing scale and which tags (line/valve/inst/equip/stru) exist

→ **Effect:** The engineer just picks files — correct tagging is auto-checked, so nothing is missed and nothing is over-drawn

3. Standardise annotation, dimension and font rules across every sheet

– Central control of tag symbols, dim colors/gaps, elevation prefix and a single unified drawing font

→ **Effect:** Every plan follows the same house style without hand-editing, cutting review comments and rework

Required inputs : One or more AVEVA E3D-exported arrangement (plan) DXF drawings

Main Window – Item Guide

General Tab

The screenshot shows the 'Piping GA Drawing Option from AVEVA E3D' dialog box. It has three tabs: 'General' (selected), 'Annotation', and 'Dimension'. The 'General' tab contains the following elements:

- Tab selector:** 'General' / 'Annotation' / 'Dimension' (1)
- Selected arrangement DXF file list:** A list box showing 'No file selected' (2)
- Select DXF drawings:** A button with a plus icon (3)
- Remove Selected / Clear All / Open:** Three buttons: 'Remove Selected', 'All' (4), and 'Open'
- Elevation Expression:** Two buttons: 'BOP' (5) and 'TOP'
- Elevation prefix:** Two buttons: 'EL.' (6) and 'GL.'
- Nozzle Table Create:** Two checkboxes: 'Nozzle Table' (7) and 'Nozz No Annotation'
- Tagging Option From E3D Draft Data:** A group of checkboxes:
 - Line No Annotation (checked)
 - Valv No Annotation (checked)
 - Inst No Annotation (checked)
 - Support Annotation (unchecked)
 - Flow Mark (checked)
 - Equip Annotation (checked)
 - Stru Annotation (checked)
 - WP EL (unchecked)
 - Annotation (8) (unchecked)
 - Tie Point Annotation (unchecked)
- All Dwg Text Font:** A dropdown menu showing 'Arial' (9)
- Buttons:** 'Apply' (10), 'Save', 'Load' (11), and 'Download'

1. Tab selector: General / Annotation / Dimension
2. Selected arrangement DXF file list
3. Select DXF drawings
4. Remove Selected / Clear All / Open
5. Elevation Expression: BOP / COP / TOP
6. Elevation prefix: EL / FL / GL
7. Nozzle Table + Nozz No Annotation checkboxes
8. Tagging Option checkboxes (auto-detected)
9. All Dwg Text Font (unify to one font)
10. Apply — run full edit chain
11. Save / Load options, Download results

Main Window – Item Guide

Annotation Tab

Piping GA Drawing Option from AVEVA E3D

Piping GA Drawing Option from AVEVA E3D

General **Annotation** Dimension

Line No Annotation

☒ Line Cyan **1** Text Yellow **2**
Text Size 2.5 Line No OverFlow Off **2**
☐ Min Line no : Left, Up/Right, Low /Right, Up/Right, Low **3**

Equip Annotation

☒ Line Cyan **4** Text Cyan **4**
Text Size 3 **4** Symbol Type Default **4**

Stru. Column No Annotation

☒ Line Cyan **5** Text Yellow **5**
Text Size 4.5 **5** Circle Size 13

Valv No Annotation

☒ Line Cyan **6** Text Yellow **6**
Text Size 2 **6** Symbol Type Rhombus **6**

Inst No Annotation

☒ Line Cyan **7** Text Yellow **7**
Text Size 2 **7** Symbol Type Circle **7**

WP EL.

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Apply Save Load Download

1. Line No: line/text color and text size
2. Line No OverFlow toggle
3. Min Line no corner-keep checkbox
4. Equip Annotation: colors, size, symbol type
5. Stru. Column No: colors, size, circle size
6. Valv No Annotation: Rhombus symbol type
7. Inst No Annotation: Circle symbol type
8. WP EL. block (scroll for more blocks)
9. Scrollbar for annotation blocks

Main Window – Item Guide

Dimension Tab

Piping GA Drawing Option from AVEVA E3D

Dimension Option

Unit: Milli Arrow end Type: Arrow

Dim. Leader Line: Dotx2 Dup. Line Dim: DN50

Piping Dimension :

☒ Line Cyan Text Yellow Gap&Size: 15 2.5

Equipment Dimension :

☒ Line Cyan Text Yellow Gap&Size: 10 2.5

Structure Dimension :

☒ Line Cyan Text Yellow Gap&Size: 10 2.5

Expression

Dimension : ☒ Left ☒ Right ☒ Up ☒ Down

Stru. Col. No ☒ Left ☒ Right ☒ Up ☒ Down

Dwg Limit ☒ L+U ☐ L+D ☐ R+U ☒ R+D

Apply Save Load Download

1. Unit + Arrow end Type selectors
2. Dim. Leader Line + Dup. Line Dim selectors
3. Piping Dimension: colors + Gap&Size
4. Equipment Dimension: colors + Gap&Size
5. Structure Dimension: colors + Gap&Size
6. Dimension sides: Left / Right / Up / Down
7. Stru. Col. No sides: Left / Right / Up / Down
8. Dwg Limit corners: L+U / L+D / R+U / R+D
9. Apply / Save / Load / Download buttons

Processing Steps

What 'Apply' runs, in order

1. Assign E3D layers
2. Create centerlines
3. Apply drawing settings
4. Edit symbols
5. Delete duplicate line-no
6. Stretch line-no leaders
7. Create dimensions
8. Edit dimensions
9. Edit tag numbers
10. Arrange tag numbers
11. Unify drawing font
12. Purge unused resources
13. Save as _ALL.DXF

Overview

What it does

1. E3D arrangement (plan) 200-series pipeline — centerline
2. Symbols
3. Line-no stretch
4. Dimensioning
5. Tag-no arrange
6. With option dialog

Category: Piping Arrangement Edit

Folder: PIPING_ARRANGEMENT_EDIT/ACE_E3D

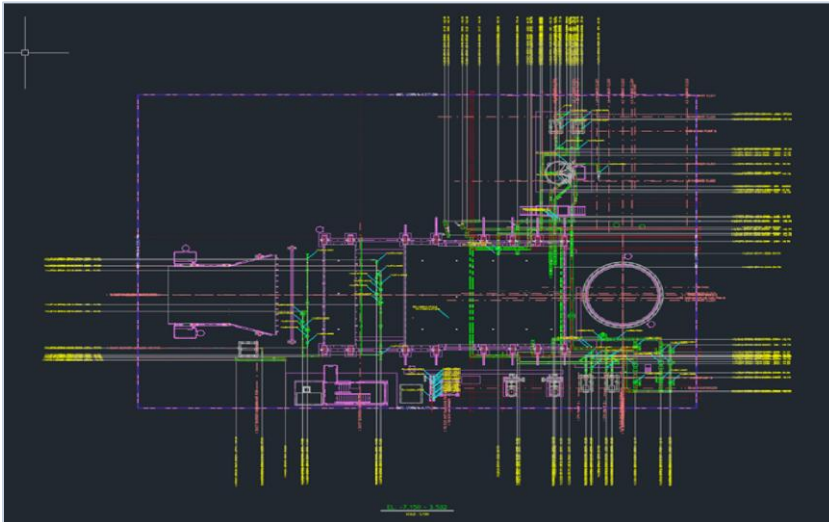
Main scripts: ACE_RUNALL.PY

Scripts / Status: 14 scripts · Operational

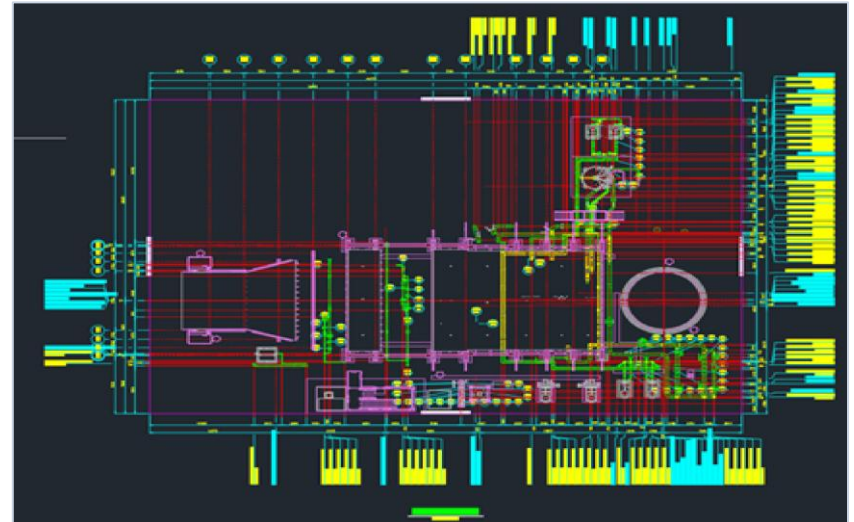
Tags: E3D · 평면도 · 200번대 · Tkinter

Output Samples

Output Samples (1/3)



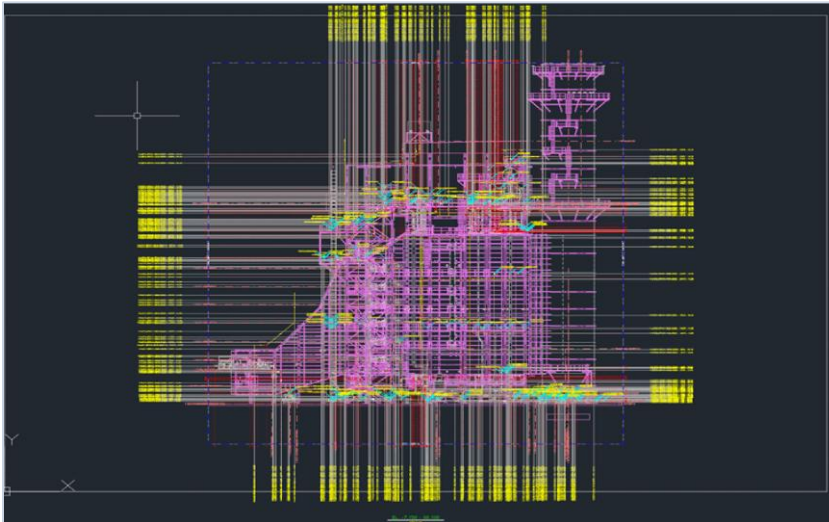
E3D Input File for Piping GA 1



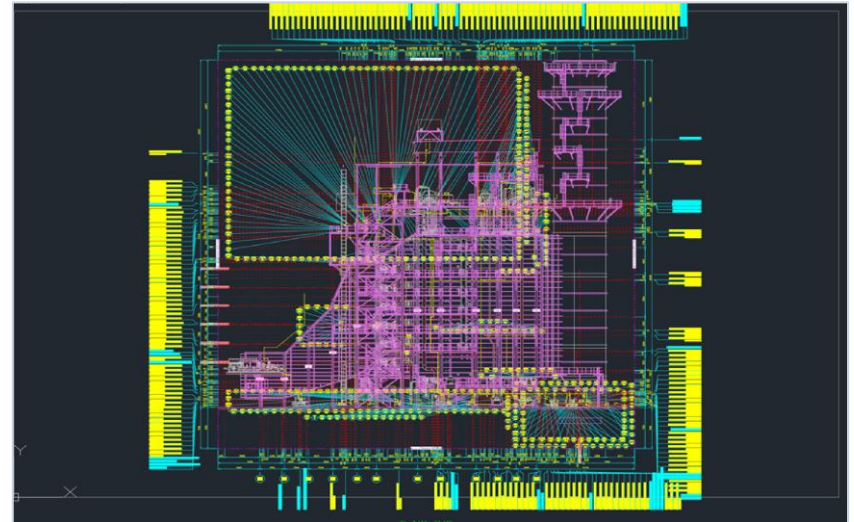
E3D Auto Output file for Piping GA 1

Output Samples

Output Samples (2/3)



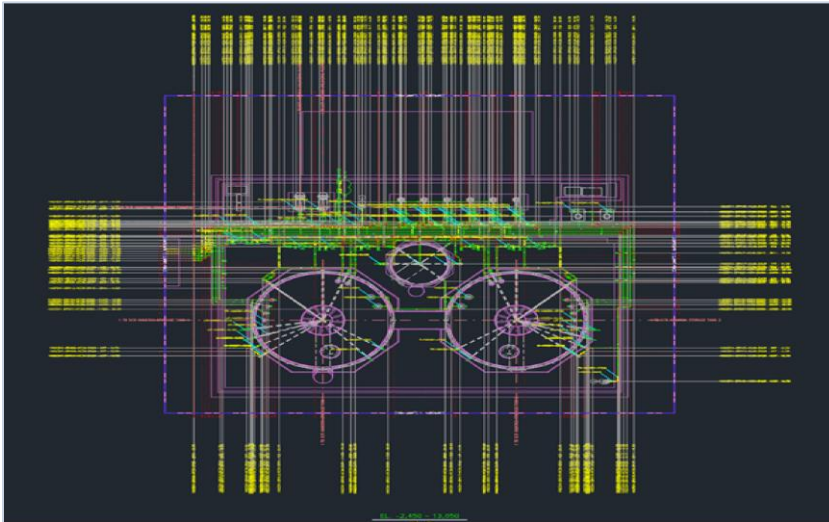
E3D Input File for Piping GA 2



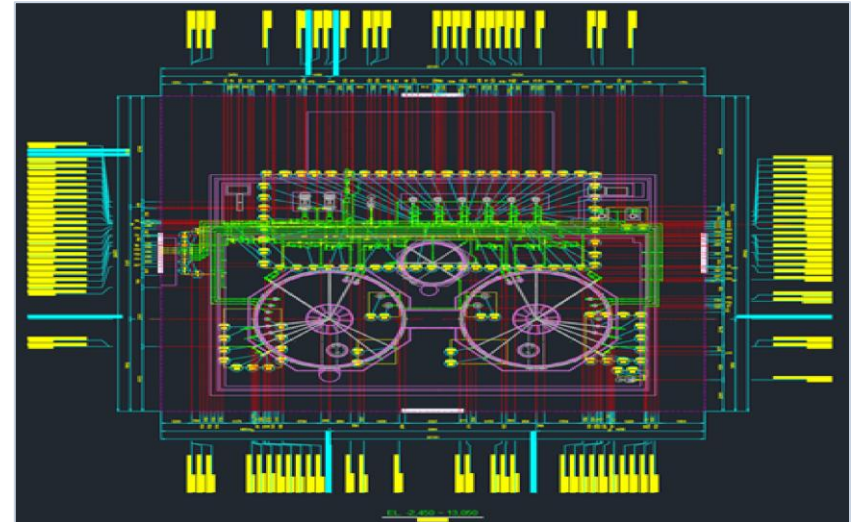
E3D Auto Output file for Piping GA 2

Output Samples

Output Samples (3/3)



E3D Input File for Piping GA 3



E3D Auto Output file for Piping GA 3

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Piping Arrangement Edit.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: ACE_RUNALL.PY.
- Output samples and the program window above reflect the current build.

Thank You

ACE Arrangement Edit (E3D) · Plant Engineering Total Service System